Technical Test — Product Engineer (Web): Octopus Website

Overview

Your task is to build a small **demo website**, not a full e-commerce site. The demo has two pages from the Octopus consumer electronics website: the **Homepage** and one **Product Detail Page** (Octopus Countertop Blender), both fully responsive, exactly as shown in the Figma flows.

- **Time available:** 5 calendar days from when you receive this brief.
- **Expected effort:** 2–3 days of work at most. If AI tools help you finish sooner, even better.
- **AI coding tools are required:** build the demo with them (for example Claude Code, Cursor or Copilot). You must still be able to explain every part of the code in the review call.
- **What we look at:** fidelity to the design across breakpoints, component structure, code quality, performance and accessibility, and how you reason about trade-offs.

What to build

The reference is the Figma file Octopus Website – Test, page "Flows". It contains two pages, each designed at three breakpoints:

Page	Desktop (≥ 1200 px)	Tablet (810–1199 px)	Mobile (< 810 px)
Homepage	Yes	Yes	Yes
Product Detail Page — Octopus Countertop Blender	Yes	Yes	Yes

The **Homepage** includes: header with navigation, hero (Octopus Ionic Hair Dryer), Best sellers carousel, category sections (Kitchen, Hair Care, Home Security, Accessories) with product cards, "Explore all categories", "Find us at" store banner, and footer.

The **Product Detail Page** includes: header, product hero with title, description, colour selector and "Where to buy" button, image gallery with feature cards, specification sections (Technical specifications, Dimensions & weight, Product information, Downloads), Best sellers carousel, and footer.

You can export images from Figma. If an asset is missing, use a neutral placeholder of the right size and ratio.

Functional requirements

Each row below is an acceptance criterion; the Figma frames show the exact layouts and states.

#	Area	Expected behaviour
1	Responsive layout	Both pages match the three Figma breakpoints and adapt fluidly between them, with no horizontal scroll.
2	Header	Desktop shows the full navigation and the "Where to buy" button. Below 1200 px the navigation collapses into a menu that opens and closes, works with the keyboard and closes on Escape.
3	Best sellers carousel	Scrolls horizontally with arrows and pagination dots on desktop, and with swipe on touch devices. Arrows are disabled at the start and end.
4	Product cards	Rendered from data, not hard-coded. "Learn more" on the Countertop Blender opens its Product Detail Page; the other cards may link to the same page.
5	Category sections	Layouts follow Figma at each breakpoint (large feature card plus smaller cards, rows becoming stacked columns on mobile).
6	Explore all categories	Category tiles link to an anchor or a placeholder route.
7	Product Detail Page hero	Shows title, description, colour selector and "Where to buy". Selecting a colour updates the selected state.
8	Specification sections	Technical specifications, Dimensions & weight, Product information and Downloads expand and collapse as an accessible accordion.
9	Footer	Links and layout match Figma at each breakpoint.

#	Area	Expected behaviour
10	Images	Responsive images with correct aspect ratios, lazy loading below the fold, and no layout shift while they load.

Technical requirements

Use a modern web framework you are comfortable with. **Next.js with React and TypeScript** is preferred; Astro, Nuxt or SvelteKit are also fine if you explain why.

- **Content from data:** products, categories and page sections come from a local JSON file or a mock API, not from markup. Adding a product to the data should be enough to show it in the right section.
- **Components:** build reusable components (product card, carousel, section header, accordion, button) and reuse them across both pages.
- **Design tokens:** colours, typography and spacing come from Figma and are defined once (CSS variables, Tailwind config or similar).
- **Browsers:** latest Chrome, Safari, Firefox and Edge, plus iOS Safari and Android Chrome.
- **Performance:** Lighthouse score of 90 or higher for Performance on the Homepage (mobile).
- **Accessibility:** semantic HTML, visible focus states, full keyboard navigation, alt text, and WCAG AA colour contrast. Lighthouse Accessibility of 90 or higher.
- **SEO basics:** page titles, meta descriptions and Open Graph tags for both pages.
- **Tests:** at least a few component tests (for example carousel and accordion) and one end-to-end test that goes from the Homepage to the Product Detail Page.
- **Visual fidelity:** follow layout, typography and spacing in Figma. Pixel-perfect is not required; consistent spacing and correct behaviour between breakpoints are.

Sample data shape, which you may adapt:

```
type Product = {
  id: string;
  slug: string;
  name: string;
  category: "kitchen" | "home-security" | "hair-care" | "accessories";
  shortDescription: string;
  image: string;
  bestSeller: boolean;
  colours?: string[];
  specs?: { title: string; items: string[] }[];
};
```

Out of scope

Do not build any of the following: cart, checkout or any purchase flow, user accounts, search, a CMS or real backend, other product or category pages, the store locator behind "Find us at", contact forms, or cookie banners. Links that point outside the two pages can go to a placeholder route.

Deliverables

Share a Git repository (GitHub or GitLab, public or with access granted to us) containing:

- [ ] Source code with a clear commit history; small, meaningful commits are preferred over a single final commit.
- [ ] A README covering setup and run instructions, the architecture and component approach, trade-offs and what you would do with more time.
- [ ] Tests that run with a single command.
- [ ] A live deployed link (Vercel, Netlify or similar).
- [ ] Lighthouse reports (mobile) for both pages, as screenshots or exported files.

In the README, describe briefly how you used AI coding tools: which tools, for which parts, and where you had to correct or rewrite their output.

Evaluation criteria

We score five areas; design fidelity and responsive behaviour carry the most weight.

Area	Weight	What we look for
Design fidelity	25%	Both pages match Figma in layout, typography, spacing and states.
Responsive behaviour	25%	Clean layouts at all three breakpoints and in between; menu, carousel and accordion work on touch and desktop.
Code quality and components	20%	Reusable components, content driven by data, typed models, design tokens defined once.
Performance and accessibility	15%	Lighthouse targets met, keyboard and screen-reader friendly, no layout shift.
Tests and docs	15%	Meaningful tests, clear README, honest trade-offs.

Polish such as subtle animations, hover states or image transitions is a bonus, never a substitute for the core requirements.

Follow-up and questions

After submission we schedule a 45-minute review call: a 10-minute demo by you, then a code walkthrough and a short discussion of how you would extend the site, for example connecting a headless CMS, adding more product pages or localisation.

If anything in the brief or the Figma file is unclear, make a reasonable assumption, note it in the README and keep going. You can also email us questions; we reply within one working day.